Supplement to "The maximum size of a blackout that keeps every lattice rectangle identifiable"

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This supplement describes the SAT encodings behind the upper bounds of the theorem on small heights, proves that they are faithful, and lists the instances, the certificates, and the formal proofs. It also proves the remark on 7 rows. We refer to results of the paper by their names. Paths are relative to the root of the repository https://github.com/apemm/rectangle-blackouts.

S1 The SAT encodings

Let $C(R)$ be the corner set of a rectangle R. A blackout S is valid if and only if it contains no set $C(R_1)\triangle C(R_2)$ with $R_1 \neq R_2$ (the lemma on presentations in Section 2 of the paper).

Encoding A has a variable for each grid point and a clause for each inclusion-minimal set of the form $C(R_1)\triangle C(R_2)$ with $R_1 \neq R_2$. The clause says that this set is not fully black.

Encoding B has such clauses only for the pairs of rectangles that share a corner. It also has a variable y_R for each rectangle R, which is forced to be true if R is fully black, and it allows at most 1 true y_R.

A *structured* instance on 6 rows fixes one row graph from the equality structure theorem (a $K_{3,3}$, for $m+9$ points) or from the lemma on 6 rows and $m+8$ points ($K_{3,3}$ minus an edge, $K_{2,4}$, or $K_{3,3}$ with 1 empty column). Each column chooses a type, which is a single row, an edge of the graph, or the empty set in the last case. Each edge is used once, and the empty type is used once in the last case. The instance keeps those clauses of encoding B that involve a tilted rectangle and whose set can be fully black for these types. The size of the blackout is fixed by a sequential counter [5] or a totalizer [1].

Lemma S1. *Let S be a blackout of the $h \times m$ grid.*

(a) S is valid if and only if it contains no inclusion-minimal set of the form $C(R_1)\triangle C(R_2)$ with $R_1 \neq R_2$.

(b) S is valid if and only if at most 1 rectangle is fully black and S contains no set $C(R_1)\triangle C(R_2)$ for distinct rectangles R_1, R_2 that share a corner.

(c) Let $h = 6$ and $m \geq 3$, and let S be valid with $m+9$ or $m+8$ points. Then for one of the row graphs above, the choice of the black rows of column i as the type of column i extends to a satisfying assignment of the structured instance.

Proof. (a) Every set $C(R_1)\triangle C(R_2)$ contains an inclusion-minimal one. (b) For rectangles without a common corner, the set $C(R_1)\triangle C(R_2)$ is $C(R_1) \cup C(R_2)$, which is black exactly when both rectangles are fully black. (c) The equality structure theorem or the lemma on $m+8$ points gives the row graph, and the black rows of every column form an allowed type. Set y_R to true exactly for the fully black rectangles. By (b) this satisfies encoding B, and the structured instance has a subset of its clauses. □

Hence the point variables of encodings A and B describe exactly the valid blackouts of the chosen size. On 6 rows, $f(6, m) \leq m + 8$ once the 10 structured instances with $m + 9$ points are unsatisfiable, and $f(6, m) \leq m + 7$ once the further 115 instances with $m + 8$ points are.

S2 Instances and certificates

CaDiCaL [2] produced an LRAT proof for every unsatisfiable instance in Table S1. Each of the 739 proofs was accepted by 2 checkers, the program `lrat-check` of drat-trim [6, 3] and the LRAT checker of the Lean core library. The proofs are large and are not stored. Their hashes, the hashes of the CNF files, and the results of both checkers are in `jis/data/unsat_certificates.json` and `jis/data/six_row_excess8_certificates.json`. The pipeline in `jis/unsat` recreates the proofs. Two separately written generators produce the same clauses for the instances of encoding A and for the 70 structured instances with $m + 9$ points. Encoding B also finds all 345 structured instances with $m + 8$ points unsatisfiable. SAT enumeration uses PySAT [4].

bound	instances	proofs
$f(5, 6) < 12$, $f(5, 7) < 13$	encoding A, 1 instance each	2
$f(6, 6) < 13$, $f(6, 7) < 14$	encoding A, 1 instance each	2
$f(6, 8) < 16$	encoding A, split by the first column	64
$f(7, 7) < 15$	encoding A, split by the first column	128
$f(7, 8) < 17$	encoding A, split by the first column	128
$f(6, m) < m + 9$ for $8 \leq m \leq 14$	structured, 10 row graphs for each width	70
$f(6, m) < m + 8$ for $9 \leq m \leq 11$	structured, 115 row graphs for each width	345

Table S1: The unsatisfiable instances.

The 72 and 1440 candidates with $m + 4$ points on the 4×4 and 4×5 grids are searched by `verify_narrow.py`, which finds 2 rectangles with the same presentation in each of them.

S3 Explicit blackouts

The blackouts and test boards of Section 6 of the paper are checked from the definition by the scripts in `jis/scripts` and by `papers/extremal/scripts/check_constructions.py`. The second script lists the tilted rectangles through their leftmost corner, compares all presentations with at most 2 points, and looks for an axis-aligned rectangle with the same presentation. It also checks the explicit construction with spacing $W_h + 1$ for $3 \leq h \leq 8$, and blackouts built as in Section 5 for $4 \leq h \leq 14$ and $h = 16$. In these, each double gets the smallest column that is not bad, where the conditions (B1), (B2), and (B3) are tested as defined. At $h = 16$, all 3 blackouts built in this way are valid, and all 5 blackouts built with (B1) alone are not valid. The script runs in about 2 minutes.

S4 Seven rows from width 21

This section proves the remark on 7 rows in Section 6 of the paper. Let T be the blackout

$$16, 36, 5, 25, 2, 2, 14, 7, 12, 27, 2, 7, 67, 7, 47, 7, 7, 56, 45, 34, 23, 1, 1, 1, 2^{21}$$

of the 7×45 grid, written as the list of the black rows of its columns, where 2^{21} stands for 21 columns black only in row 2. Its 12 doubles lie in the first 21 columns and carry the 12

pairs of an odd and an even row. For $m \geq 21$ let T_m be the blackout of the $7 \times m$ grid that agrees with T on the columns 1 to 45 and whose later columns are black only in row 2. The script `papers/extremal/scripts/check_constructions.py` checks that T is valid. Since $f(7, m) \leq m + 12$ by the row bound, the next lemma gives $f(7, m) = m + 12$ for every $m \geq 21$.

Lemma S2. *If T is valid, then T_m is valid for every $m \geq 21$.*

Proof. For $m \leq 45$ the blackout T_m is the part of T in its first m columns, and every rectangle of the smaller grid has the same presentation in T. Let $m > 45$, and suppose that distinct rectangles R_1, R_2 have the same presentation in T_m. Then the corners that belong to just one of them are black. We show that both rectangles lie in the columns 1 to 45, which contradicts the validity of T. A tilted rectangle on 7 rows meets at most 11 consecutive columns (the span bound with $W_7 = 10$). The doubles have distinct pairs, which form a bipartite graph. So the row-column graph has no cycle of length 4 or 6. Hence R_1, R_2 are not both axis-aligned, and no axis-aligned rectangle is fully black. A tilted rectangle has at most 2 corners in one row. So a tilted rectangle with 3 or more black corners has a corner in a column $x \leq 24$, and it lies in the columns 1 to 34.

Suppose that R_1, R_2 have at most 1 common corner. Then both have 3 or more black corners. A tilted one lies in the columns 1 to 34. If R_2 is axis-aligned, then it is not fully black, so it has a common corner p with the tilted rectangle R_1, and its other column contains 2 black points. This column is a double, so R_2 also lies in the columns 1 to 34.

Suppose that R_1, R_2 have 2 common corners p, q, and that pq is a side of both or a diagonal of both. By the lemma on such pairs in Section 2 of the paper, their other corners form a fully black rectangle R_3. It is tilted, so it lies in the columns 1 to 34. A tilted R_i shares 2 corners with R_3, so it lies in the columns 1 to 44. If R_i is axis-aligned, then the other rectangle is tilted and has no horizontal or vertical side. So pq is a diagonal of R_i, and the columns of R_i are those of its 2 corners in R_3.

In the remaining case pq is a side of one rectangle, say R_1, and a diagonal of R_2. Let AB be the side of R_1 opposite to pq, and let CD be the other diagonal of R_2. The points A, B, C, D are black. Suppose first that AB is not vertical. Then both rectangles lie in 21 consecutive columns. Indeed, if both are tilted, they share the corner p and each meets at most 11 consecutive columns. If one of them is axis-aligned, its columns are those of p and q, which are columns of the other rectangle (compare Section 5.1 of the paper). If one of A, B, C, D lies in a column $x \leq 24$, then both rectangles lie in the columns 1 to 44. Otherwise A, B, C, D lie in row 2. Then pq is parallel to AB and horizontal, and the midpoint of CD, which is the midpoint of pq, lies in row 2. So p, q lie in row 2 as well, and R_1 has no area, a contradiction. Suppose now that AB is vertical. Then A, B are the 2 points of a double, R_1 is axis-aligned, and the rows of p, q are an odd and an even row. The rectangle R_2 is tilted with the vertical diagonal pq and the black diagonal CD. If C or D lies in a column $x \leq 24$, then R_2 and R_1 lie in the columns 1 to 34. Otherwise C and D lie in row 2. Then the rows of p and q are 1 and 3, a contradiction. □

S5 Formal proofs

The Lean files are in `lean/KageyProblems/Problem001`. They contain formal proofs of the lemma on presentations, of the lemma that 3 corners determine a rectangle, of the span bound, of the row bound, of the finite collar theorem, and of the alternating seas proposition under the assumption that the doubles are $2W_h + 2$ columns apart (`GeneralLocal.lean` and `GeneralTheorem.lean`). This gives $f(h, m) = m + \lfloor h^2/4 \rfloor$ for every $h \geq 3$ and $m \geq q_h + (q_h - 1)(2W_h + 1)$. Validity tests of S_0, S_8, the 6×8 and 6×15 blackouts, and the 5×13

and 6×23 test boards are checked by the Lean kernel. Together with the formal row bound and collar theorem, this proves $f(5, m) = m + 6$ for $m \geq 8$, $f(6, m) = m + 9$ for $m \geq 15$, $f(5, 5) \geq 10$, and $f(6, 8) \geq 15$ in Lean.

Not formalized are the spacing $W_h + 1$, the threshold theorem and its corollary, the equality structure theorem, the lemma on 6 rows and $m + 8$ points, the blackouts on 7 rows, and Lemmas S1 and S2.

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